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ORIGINAL PAPER

Outlier SNPs detect weak regional structure against a background of genetic homogeneity in the Eastern Rock Lobster, *Sagmariasus verreauxi*

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Abstract

Genetic differentiation is characteristically weak in marine species making assessments of population connectivity and structure difficult. However, the advent of genomic methods has increased genetic resolution, enabling studies to detect weak, but significant population differentiation within marine species. With an increasing number of studies employing high resolution genome-wide techniques, we are realising that the connectivity of marine populations is often complex and quantifying this complexity can provide an understanding of the processes shaping marine species genetic structure and to inform long-term, sustainable management strategies. This study aims to assess the genetic structure, connectivity, and local adaptation of the Eastern Rock Lobster (*Sagmariasus verreauxi),* which has a maximum pelagic larval duration of 12 months and inhabits both subtropical and temperate environments. We used 645 neutral and 15 outlier SNPs to genotype lobsters collected from the only two known breeding populations and a third episodic population—encompassing *S. verreauxi*'s known range. Through examination of the neutral SNP panel, we detected genetic homogeneity across the three regions, which extended across the Tasman Sea encompassing both Australian and New Zealand populations. We discuss differences in neutral genetic signature of *S. verreauxi* and a closely related, co-distributed rock lobster, *Jasus edwardsii*, determining a regional pattern of genetic disparity between the species, which have largely similar life histories. Examination of the outlier SNP panel detected weak genetic differentiation between the three regions. Outlier SNPs showed promise in assigning individuals to their sampling origin and may prove useful as a management tool for species exhibiting genetic homogeneity.

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Introduction

Researchers and resource managers traditionally believed that genetic homogeneity was the only genetic structure possible for a marine species with a pelagic larval duration (PLD) in the order of months to years. As the ocean has few physical barriers, species with longer PLDs have the potential to disperse over larger geographical scales (Shanks et al. 2003; Siegel et al. 2003), enhancing gene flow between geographically isolated populations. Early population genetic studies, limited to a small number of neutrally evolving genetic markers, often lacked statistical power to detect population divergence (Ouborg et al. 1999; Allendorf 2017) characteristic of many marine species (Allendorf et al. 2010), reinforcing the assumption of high connectivity between populations in the marine environment. However, with the advent of powerful genomic techniques, large panels of SingleNucleotide Polymorphisms (SNPs) have increased genetic resolution, enabling studies to detect weak, but significant population differentiation within marine species (Benestan et al. 2015; Araneda et al. 2016; Villacorta-Rath et al. 2016).

Using high resolution genomic methods, such as RAD-Seq (Baird et al. 2008; Peterson et al. 2012) and DArTSeq (Sansaloni et al. 2011), enables analyses to differentiate between neutrally evolving SNPs and "outlier" SNPs—loci with higher genetic divergence than expected under neutral conditions (Perez-Figueroa et al. 2010). The higher genetic divergence detected in these loci is potentially caused by environmental selection; however, it should be noted that outlier loci may also be detected due to Type I error or demographic processes, such as range expansions and pre- or post-zygotic isolation (Bierne et al. 2011). Nevertheless, outlier loci are often more powerful than neutral SNPs in delineating genetic structure and reliably assigning individuals to sampling regions in species exhibiting genetic structure (Freamo et al. 2011; Milano et al. 2014; Araneda et al. 2016; Villacorta-Rath et al. 2016). Furthermore, if outlier SNPs can be linked to biological function, they may provide useful insight into the ecological drivers of adaptation within their environment.

With an increasing number of studies employing high resolution genome-wide techniques, we are realising that the connectivity of marine populations is often complex: influenced by a multitude of factors including local oceanographic features, life-history factors, and historical environmental changes (Zardoya et al. 2004). Species that occupies the same geographical location, is closely related, and/or has similar life-history traits can exhibit markedly different patterns of genetic structure (Ovenden et al. 2013; Holland et al. 2017; Momigliano et al. 2017), indicating that the traditional assumption of high population connectivity is inappropriate for marine species. Understanding and quantifying the complexity of connectivity in the marine environment is important on an evolutionary scale to understand processes that shape the genetic structure in marine species, but also on a contemporary scale to inform long-term, sustainable conservation and management strategies (Hauser and Carvalho 2008; Ovenden et al. 2013; Bernatchez et al. 2017).

Population structure and connectivity in the Eastern Rock Lobster, *Sagmariasus verreauxi*, is currently poorly understood. This species supports commercial fisheries in New South Wales (NSW), Australia (NSW Department of Primary Industries 2007) and in the North Island of New Zealand (NZ; Kensler and Skrzynski 1970). Separate management strategies exist for both countries based on the results of an early genetic assessment, which used six mitochondrial regions, indicating separate genetic populations in NSW and NZ (Brasher et al. 1992). While *S. verreauxi* are yet to be assessed with high resolution genetic markers, SNPs recently confirmed a trans-Tasman genetic break in the co-distributed, closely related, commercially important Southern Rock Lobster, *Jasus edwardsii* (Villacorta-Rath et al. 2016). Oceanographic models indicate that trans-Tasman gene flow for both species of lobster is possible due to their long PLDs (Chiswell et al. 2003); microsatellite studies have indicated that for *J. edwardsii*, trans-Tasman gene flow occurs at levels low enough to maintain genetic structure (Thomas and Bell 2013). As *S. verreauxi* and *J. edwardsii* have many similar life-history characteristics, such as long pelagic larval durations, high fecundity, and restricted breeding times (Table 1), we could expect the subdivision detected for *J. edwardsii* to be mirrored in *S. verreauxi* populations. However, these lobster species do exhibit some differences in their life history, such as geographically restricted vs unrestricted breeding locations and large vs small migratory movements (Table 1), which can result in disparate patterns of genetic structure (Ovenden et al. 2009). Comparing and contrasting the genetic structure of closely related, co-distributed species provide insights into the factors responsible for shaping genetic structure in the marine environment (Kelly and Palumbi 2010).

Sagmariasus verreauxi inhabits a wide latitudinal range, covering subtropical to temperate environments. The two geographically restricted breeding grounds occur in the subtropical environments of northern New South Wales, eastern Australia (Montgomery 1992) and Cape Reinga, New Zealand (Booth 1986), while a third small, non-breeding population occurs in temperate Tasmania, the southernmost area of Australia (Montgomery et al. 2009; Fig. 1). Distribution across heterogeneous environments, together with limited breeding sites, suggests that spatially varying selection may play a role in shaping the adaptive genetic structure of *S. verreauxi*.

Table 1 Comparison of the life-history characteristics of two rock lobster species, *Jasus edwardsii* and *Sagmariasus verreauxi*

	Sagmariasus verreauxi	*Jasus edwardsii*
Physical separation between Australian and NZ populations	~10 million years ago (George 1997)	~10 million years ago (George 1997)
Distribution	Inhabits coasts of New South Wales, Australia and north east coast of North Island, New Zealand, with a small, episodic population in Tasmania, Australia (Kensler and Skrzynski 1970; Montgomery et al. 2009)	Inhabits all coasts of Southern Australia (including Western Australia, South Australia, Victoria, Tasmania) and New Zealand (Booth 2006)
Breeding locations	Restricted breeding grounds occur along coast of northern New South Wales, Australia and Cape Reinga, New Zealand (Booth 1984; Montgomery 1992)	Breeding occurs in all areas of distribution (Booth and Breen 1994; Booth and Phillips 1994)
Breeding timing	Breeding occurs once/annum between October–January (Booth 1984). Eggs hatch between December to January (Kensler 1967; Booth 1979)	Breeding occurs once/year in June to September (George 2005). Eggs hatch between September and October (Booth and Stewart 1992)
Fecundity	385,000–2 million eggs per female (Kensler 1967).	43,918–660,156 eggs per female (Green et al. 2009)
Pelagic larval duration	8–12 months (Booth 1986; Montgomery and Kittaka 1994; Montgomery and Craig 2005)	Up to 24 months (Booth and Phillips 1994)
Migration	Recruits settle to the south of the breeding grounds before undergoing a countercurrent migration at the onset of maturity to join the breeding population. Migration distance ranges from 200 to 1070 km (Booth 1986)	Majority of migrations are small on- and off-shore movements < 100 km (McKoy 1983; Linnane et al. 2005)
Genetic structure	Australian and New Zealand populations consist of a single genetic stock, with outlier SNPs indicating weak regional differences	Australian and New Zealand populations and populations within Australia are genetically distinct (Morgan et al. 2013; Thomas and Bell 2013; Villacorta-Rath et al. 2016)

Sagmariasus verreauxi has been caught in Tasmanian waters since 1886 (Saville-Kent 1886), as bycatch of the *J. edwardsii* focussed Tasmanian Rock Lobster Fishery. Tasmanian *S. verreauxi* are currently rare and recruitment into the population is irregular. Recruits are believed to originate in NSW waters, and are transported to Tasmania by the southerly flowing East Australian Current (EAC; Gardener et al. 2000; Pecl et al. 2009). However, evidence for this is anecdotal and originates from the observation of a large number of *S. verreauxi* recruits settling on Tasmania's east coast, coinciding with the occurrence of a strong EAC (Gardener et al. 2000). Importantly, no records exist of egg-bearing females occurring within Tasmanian waters (Pecl et al. 2009), despite intensive fishing and monitoring practices in the area. Due to predictions of climate change induced EAC intensification (Cai et al. 2005; Ridgway and Hill 2012), there is some concern that this species may undergo a population expansion and establish a breeding population in Tasmania in the future, which may have detrimental effects on local *J. edwardsii* populations (Pecl et al. 2009; Robinson et al. 2015). Thus, it is important to clarify the origins of the Tasmanian recruits to gain an understanding of the likely response of this peripheral population to intensification of the southward EAC penetration in the future.

In this study, we used double digest restriction-site associated DNA sequencing (ddRADSeq) to identify neutral and outlier SNPs to assess the genetic structure, connectivity, and local adaptation between *S. verreauxi* populations. Specifically, we aimed to clarify whether NSW and NZ *S. verreauxi* populations consist of a single panmictic stock or mirror the genetic structure detected in the closely related *J. edwardsii*. Dependant on the genetic structure detected, we compared and contrasted the times of population divergence and life history of both rock lobster species in an attempt to understand the biological factors shaping their genetic structure. We also aimed to determine whether evidence of self-recruitment exists for the Tasmanian population and, if not, the most likely source of Tasmanian recruits.

Materials and methods

Tissue collections

A total of 90 *Sagmariasus verreauxi* tissue samples were obtained from three regions (NSW, NZ, and TAS) spanning the species range in 2013 and 2014 (Table 2; Fig. 1). NSW Department of Primary Industries spawning stock and commercial catch surveys provided samples from NSW and the NZ Rock Lobster Industry Council provided samples from NZ. There is no commercial *S. verreauxi* fishery in Tasmania; TAS samples were opportunistically sourced from both fishers and Stanley Seaquarium (a processing facility retaining *S. verreauxi* bycatch in captivity). Tissue samples were taken from either the telson, pleopod, or leg and stored in 90% ethanol at − 80 °C.

Molecular techniques

Total genomic DNA was extracted from approximately 20 mg of tissue using a DNeasy Blood and Tissue Kit (Qiagen). DNA samples were divided into two groups for library preparation, with each group including three intra- and three inter-library replicates. Replicate pairs were sourced from a single DNA extraction, but were treated as separate samples during library preparation. Libraries were processed using a modified version of Peterson et al. (2012) ddRADSeq protocol (Supplementary Appendix 1). Libraries were sequenced as 100 base pairs (bp) single end (SE) reads on two lanes of an Illumina HiSeq 2500 by the Australian Genome Research Facility (AGRF; Melbourne, Australia).

Preliminary quality control of raw sequencing data

Raw read quality was assessed with FASTQC v0.10.1 (Babraham Bioinformatics 2007). Human and bacterial contaminants were identified using KRAKEN v0.10.4 (Wood and Salzberg 2014) and subsequently removed. TRIMMOMATIC v0.32 (Bolger et al. 2014) was used to remove reads with an average quality score < Phred 28. Reads matching mitochondrial DNA (mtDNA) were removed with a custom KRAKEN database consisting of *S. verreauxi* mtDNA (Doyle et al. 2015). The remaining data were demultiplexed and trimmed to 75 bp using 'process_radtags' from STACKS v1.29 (Catchen et al. 2011).

De novo assembly and SNP calling

De novo assembly was conducted using a custom pipeline, RAD-LOCI (Villacorta-Rath et al. 2016, 2018; available at https://github.com/molecularbiodiversity/rad-loci), which was developed to deal with paralogous loci retained by other available software. To include sequences with low coverage within individuals but high coverage across the data set, demultiplexed reads from all samples were initially clustered at 100% similarity, with a minimum cluster depth of 75 sequences—allowing two-thirds of the samples to be represented by at least one read. Reference loci were collated by merging clusters together—allowing a mismatch of three nucleotides—and removing monomorphic clusters (those with less than two variants) and potentially paralogous clusters (those with more than 16 variants). The upper threshold of 16 was chosen as a 3-bp mismatch with four possible nucleotide substitutions equates to 64 possible

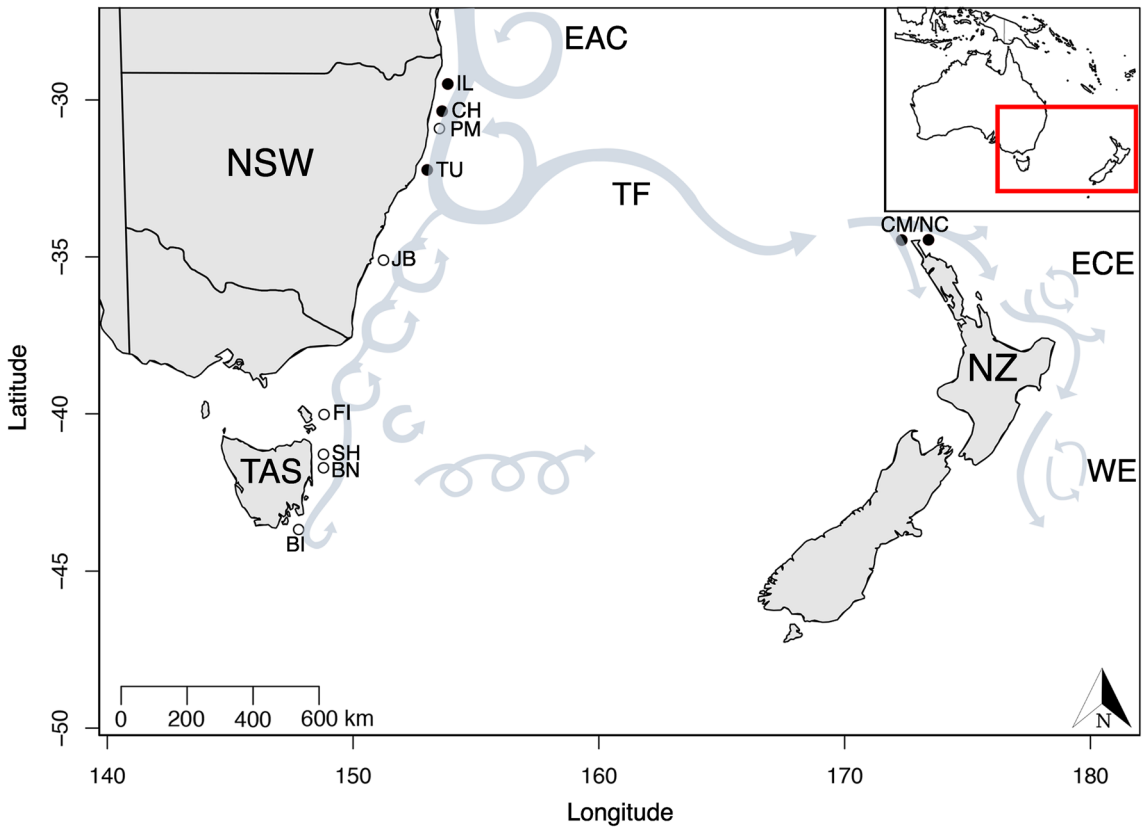

Fig. 1 Locations of *Sagmariasus verreauxi* sampling sites (listed in Table 2) within Australia and New Zealand during 2013 and 2014 sampling periods. Black dots denote sampling sites, where only adults were collected. White dots denote sampling sites, where juveniles were collected. The major local oceanographic currents depicted on the map: *EAC* East Australian Current, *TF* Tasman Front, *ECE* East Cape Eddy, *WA* Wairarapa Eddy. The study region is illustrated in a broader context in the inset map (red box)

Table 2 Sampling and sequencing information

Region	Site code	Site name	Year	No. samples collected	No. of sequenced individuals:		
					Juveniles	Adults	Unknown
NSW	IL	Iluka	2014	6	–	6	–
	CH	Coffs Harbour	2014	8	1	7	–
	PM	Port Macquarie	2013	7	6	–	–
	TU	Tuncurry	2014	2	1	1	–
	JB	Jervis Bay	2013	7	7	–	–
TAS	FI	Flinders Island	2014	3	1	–	–
	SH	Saint Helens	2014	2	2	–	–
	BN	Bicheno	2013	8	8	–	–
	BI	Bruny Island	2014	1	1	–	–
	SSA	SS Aquarium	2014	16	4	1	–
NZ	CM	Cape Maria van Diemen to North Cape	2013	19	–	3	16
	NC		2014	7	–	4	–

sequence permutations (e.g., 4^3); however, during pipeline development, the highest number of cluster members identified was 16, and hence, the upper threshold was set at 16. This step was repeated, using the same thresholds, and clusters containing variants previously associated with a different cluster in the previous filtering round were removed. Variants that mapped to a different cluster in the second filtering round were likely from repetitive or closely related regions. All samples were mapped back to the remaining loci and final filtering removed: (1) loci with > 20% missing data and (2) loci for which a single sample mapped to more than two variants, indicating a repetitive region. Remaining

loci were recorded into a RAD-tag catalog and were used as reference sequences to align the filtered sample data with VSEARCH v1.1.3 (https://github.com/torognes/vsearch) in preparation for SNP calling.

The genotype likelihoods of putative variant sites were determined using the mpileup command in SAMTOOLS v1.2 (Li et al. 2009), and variants were called using BCFTOOLS v1.2 (Li 2011) with a minimum depth of three reads. To reduce spurious genotyping calls due to low read depth, a likelihood ratio was calculated from the most- and second- most likely Phred scaled likelihood for each genotype. Samples with a ratio of ≤10 (equating to 10% error on a Phred scale) were marked as missing data.

VCFTOOLS v0.1.13 (Danecek et al. 2011) was used for subsequent filtering of variant sites. SNPs which exhibited a significant departure ($P < 0.001$) from Hardy–Weinberg Equilibrium (HWE), indels, loci with > 30% missing genotypes, and a global minor allele frequency of < 0.05 were excluded from the data set. Samples with more than 30% missing data across all loci or with an initial sequence read count of less than 1 million were excluded from further analysis. To negate high levels of site linkage within the data set, only one SNP per locus was retained using the thin option in VCFTOOLS.

Analysis of replicates and batch effects

To assess batch effects, a principal component analysis (PCA) was conducted on the full SNP panel with the intra- and inter-library replicates using the R-package *adegenet* v2.0.0 (Jombart 2008). Locus and allele error rates were calculated for all replicates (Mastretta-Yanes et al. 2015). The replicate from each pair with the highest amount of missing data was removed from further analyses. PCAs were also conducted on the full SNP panel to test for batch effects between libraries, sampling years and juveniles and adults.

Detection of markers under selection

The full SNP panel was screened for regional outlier loci using a consensus approach to reduce detection of false positives. Two programs, LOSITAN (Beaumont and Nichols 1996; Antao et al. 2008) and ARLEQUIN v3.5 (Excoffier et al. 2005), were run three times each and SNPs identified as divergent in all runs were classified as putatively positively selected. Both programs use a coalescent-based simulation to identify outlier markers by assessing the relationship between F_{ST} and expected heterozygosity, with LOSITAN assuming a neutral island model of migration. Under the assumption of an island migration model, high numbers of false positives can be detected when genetic structure is hierarchical in reality (Excoffier et al. 2009). To mitigate this, ARLEQUIN was run under a hierarchical model, with default settings of 100 demes per ten groups, 50,000 permutations and a finite island model. LOSITAN runs were performed with 50,000 simulations, neutral and forced F_{ST}, a confidence interval of 0.99 and a false discovery rate of 0.1. SNPs classified as putatively neutral or putatively positive outliers were separated into corresponding SNP panels.

To determine if outlier SNPs were associated with biological functions, a homology search of the 75-bp RAD tags containing the outlier SNPs was performed against the NCBI nr public database using BLASTn (Altschul et al. 1990) and against the American lobster, *Homarus americanus*, transcriptome (McGrath et al. 2016). A minimum E value of 10^{-6} was the threshold for a significant hit.

Genetic variation

Observed heterozygosity (H_O), unbiased expected heterozygosity (H_{UE}), and unbiased F_{ST} estimator θ (Weir and Cockerham 1984) were calculated for the neutral SNP panel using GENODIVE v2.0b27 (Meirmans and Van Tienderen 2004). Significant departures in H_O and H_{UE} among the three regions were tested using the OSx-statistic (Goudet 1995) with 10,000 permutations. Inbreeding coefficient (F_{IS}) was calculated in ARLEQUIN v3.5 (Excoffier et al. 2005). Significance values for F_{ST} and F_{IS} were based on 10,000 permutations.

Population structure and assignment tests

Regional genetic structure was examined in the neutral SNP panel using both a Bayesian clustering method, performed in STRUCTURE v2.3.4 (Pritchard et al. 2000) and Discriminant Analysis of Principal Components (DAPC), performed using the R-package *adegenet* (Jombart 2008). Both methods were used to identify the most likely number of genetically distinct clusters (K) and to assign individuals to those populations. For the STRUCTURE analysis, 20 independent runs for each value of K (1–6) were performed, under the assumption of an admixture model and allele frequencies being correlated among populations. Each run had a burn-in period of 100,000 followed by 100,000 data iterations. To evaluate the best value of K, the mean natural logarithm of the probability of the data ($\ln \Pr(X|K)$) and the ΔK method (Evanno et al. 2005) was assessed via STRUCTURE HARVESTER (Earl and vonHoldt 2012). Results of individual population assignments were visualised with CLUMPAK (Kopelman et al. 2015).

Unlike STRUCTURE, DAPC is a multivariate analysis that does not assume an underlying population genetic model (Jombart et al. 2010) and was, therefore, used to assess both the neutral and outlier SNP panels. The premise of DAPC is to minimise within group variation while

maximising variation between the groups. To determine the optimal value of K, Bayesian Information Criterion (BIC) was implemented in *find.clusters*, without sampling information. Overfitting of discriminant functions can occur when too many principal components (PCs) are retained. To avoid overfitting, a cross-validation approach was used to determine the optimal number of PCs to retain, using 75% of the samples as a training set.

To establish whether source population(s) for TAS recruits could be identified, assignment tests were performed on the neutral SNP panel using a Bayesian approach (Rannala and Mountain 1997) in GENECLASS2 (Piry et al. 2004). NSW, NZ, and approximately half of the TAS samples ($n = 9$), chosen at random, served as reference populations and the remaining TAS samples ($n = 8$) were assigned or excluded. The probability computations were based on the resampling algorithm of Paetkau et al. (2004), recommended for detecting first generation migrants (Piry et al. 2004), with 10,000 simulated individuals and an α of 0.01, as recommended by Paetkau et al. (2004).

In a number of marine population genetic studies, outlier SNPs have been shown to produce similar patterns of genetic structure as neutral SNPs, but with higher statistical power (Freamo et al. 2011; Milano et al. 2014; Araneda et al. 2016; Villacorta-Rath et al. 2016). Therefore, we examined whether outlier SNPs were able to successfully assign individuals to their sampling region. The same parameters as described for assigning Tasmanian recruits were used, with the additional likelihood of an individual genotype occurring within the population of collection (L_home). The results of the self-assignment test for the outlier SNPs were cross-validated by splitting the samples randomly into reference populations and individuals for assignment. Three different criteria were assessed for successful assignment: (a) the highest probability of occurrence; (b) 99% probability of occurrence; and (c) 95% probability of occurrence.

Allele sharing distance methods

Allele sharing distances have proven effective in detecting genetic partitioning between regions (Bowcock et al. 1994) and subpopulations (Estoup et al. 1995), when allele frequencies indicate little to no genetic differentiation. To assess subtle population genetic structure undetectable through allele frequency methods, an allele sharing distance (ASD) matrix (Bowcock et al. 1994) was created utilising the neutral and outlier SNP panels in the R-package *adegenet*. A Neighbour Joining (NJ) dendrogram was constructed from the ASD matrix with the R-package *ape* v5.0 (Paradis et al. 2004).

To test whether alleles were randomly distributed across geographical space, the neutral SNP panel was assessed using a Spatial Analysis of Shared Alleles implemented in SAShA 2 (Kelly et al. 2010). Detection of non-random allele distribution can signify a departure from panmixia. Alleles can either be under-represented in space, indicating subtle genetic partitioning, or over-distributed, suggesting that alleles occur further apart than would be expected under a panmictic model. The observed mean allele distribution (OM) was tested for significant departure of the expected mean allele distribution under panmixia (EM), with 1,000 permutations.

Gene flow analysis

The direction of gene flow between the breeding grounds in NSW and NZ was tested using the Bayesian strategy implemented in MIGRATE-N v3.6.11 (Beerli 1998; Beerli and Felsenstein 1999, 2001). In order for the MIGRATE-N analysis to be computationally tractable, a subset of 50 randomly chosen 75-bp loci were tested under the sequence model with constant mutation rates. A series of initial full migration model simulations were conducted to ensure that convergence was obtained. The full migration model described NSW and NZ as genetically distinct populations with bi-directional gene flow. Once replicate runs converged, three additional gene flow models were tested, which included NSW and NZ as genetically distinct populations with asymmetrical gene flow originating from NSW or from NZ and a model describing NSW and NZ as a single genetic population (Table 5). Each model was performed with the parameters set to one long chain with 10,000 genealogies sampled and runs concurrently five times. Sampling increment and burn-in was 100 and 100,000, respectively. To determine the most likely migration model, log Bayes Factors (LBF) were calculated from the Bezier approximation score and compared for all four gene flow models.

To examine the directionality and magnitude of contemporary migration and identify possible source populations for Tasmanian recruits, all three sampling sites were assessed using the *divMigrate* function (Sundqvist et al. 2016) from the R-package *diveRsity* (Keenan et al. 2013), employing both Nei's G_{ST} and Jost's D. Relative levels of migration are calculated in *divMigrate* through assessing the genetic differentiation between two populations and a hypothetical pool of migrants. Source and sink populations can be identified when relative migration rates are larger in one direction than the other direction (Sundqvist et al. 2016).

Results

Sequence filtering and SNP discovery

A total of 417,216,775 SE reads were obtained from sequencing 90 individuals with 9 replicated individuals

($n = 99$). Following filtering of raw reads to remove contaminant sequences, poor quality reads, and mtDNA-derived reads, the demultiplexed mean read abundance was 3.5 million reads per sample (± 1.4 million SD; range = 320,694–1,913,298). These reads were used to create the RAD-tag catalog containing 2,635 RAD-LOCI for SNP calling, from which a total of 3,384 SNPs were identified. After further site- and individual-based filtering, the final full SNP panel contained 756 SNPs, with an average read depth of 16 (± 6.7 SD). Under the hierarchical model, ARLEQUIN was unable to complete the outlier analysis as the F-statistics were negative, indicating no hierarchical structure in the data. Consensus outlier analyses of LOSITAN and ARLEQUIN finite island model detected 15 outlier SNPs exhibiting putative positive selection. From the 741 remaining SNPs, 81 were detected to be subject to balancing selection and an additional 15 SNPs had been identified as potential outliers, but did not fulfil the requirements of the consensus approach. 645 SNPs were retained in the neutral SNP panel. No significant nucleotide or transcriptome BLAST hits were returned for RAD-tags containing outlier SNPs. A total of 23 individuals, with > 30% missing data across all loci, were removed from the data set, along with one of the replicate samples, resulting in a final sample size of 29, 23, and 17 individuals for NSW, NZ, and TAS, respectively (Table 3).

Batch effect and replicate analysis

No batch effect from library preparation or sequencing was evident in the PCA of the inter-library replicates, which showed no separation of replicates across the libraries (Supplementary Fig. S1). Consistent with this result, the mean locus and allele error rates did not differ between the inter-library replicates. Non-concordant genotyping errors (e.g., different genotyping calls) were present in low abundance: occurring in 9% of SNPs.

In addition, genetic structure was not attributable to the collection of samples across two sampling years or the different life stages collected. No clear separation between the sampling years, or life stages was evident in the respective PCAs (Supplementary Fig. S2a and S2b, respectively).

Genetic variation

A founder effect (i.e., a reduction in genetic diversity), characteristic of breeding populations recently colonised by a small number of individuals, was not detected for the Tasmaninan population (Table 3). Comparison of H_O and H_{UE} between the NSW, NZ, and TAS indicated that genetic diversity did not significantly differ between regions (H_O OSx = 0.041, $P = 1.00$; H_{UE} OSx = 0.013, $P = 1.00$). Gene flow between the three regions is evidenced by the lack of regional private alleles detected and the inbreeding coefficient (F_{IS}) not significantly deviating from zero in any region.

Table 3 Genetic diversity indices for all sampling regions

Region	Neutral SNP panel			
	N	H_O	H_{UE}	F_{IS}
NSW	29	0.314	0.300	−0.160
NZ	23	0.347	0.309	−0.309
TAS	17	0.326	0.308	−0.259

No F_{IS} values significantly differed from zero

No significant differences were detected for H_O or H_{UE} among the three regions

N the number of individuals genotyped, H_O observed heterozygosity, H_{UE} unbiased expected heterozygosity, F_{IS} Wright's inbreeding coefficient

Population structuring and assignment based on allele frequency methods

No genetic differentiation was detected between regions in any of the analyses performed on the neutral SNP panel. All pairwise F_{ST} values were non-significant, indicating a lack of genetic differentiation between regions (Supplementary Table S1). The mean ln Pr($X|K$) method indicated that the optimal value of K, signified by the point of plateau, ranged from one to three genetic clusters (Supplementary Figure S3a), while the Evanno method (which is unable to detect a lack of genetic structure, e.g., $K = 1$) determined the optimal value of K to be five genetic clusters, signified by the highest ΔK (Supplementary Fig. S3b). However, visualisation of the STRUCTURE output at two, three, and five genetic clusters indicates no genetic differentiation between the sampling locations, with all individuals showing similar degrees of admixture of the genetic clusters (Supplementary Fig. S3c). The presence of a single genetic stock ($K = 1$) was indicated in the *find.cluster* results as the smallest BIC located was at $K = 1$ (Supplementary Fig. S4a). DAPC can only be computed with a minimum $K = 2$; therefore, a DAPC was not conducted for the neutral SNP panel. TAS recruits were equally highly likely to be assigned to any one of the three regions (NSW probability = 1.00; NZ probability = 1.00; TAS probability = 1.00), based on the neutral SNP panel assignment test. This indicates that the regional neutral genetic signatures cannot be distinguished from each other, which is consistent with the lack of genetic differentiation detected with F_{ST} and BIC.

Weak genetic differentiation among the three regions was detected in the analysis of the outlier SNP panel. While the number of genetic clusters was unable to be defined, BIC

indicated that the most appropriate number of genetic clusters ranged between 3 and 5, evidenced by the bend in BIC (Supplementary Fig. S4b). The outlier SNPs did not hold enough signal to determine the optimal value of K; however, visualisation of each best K with DAPC demonstrated that genetic clusters corresponded to the three geographical regions in $K=3$ and 4, albeit with some admixture present. The best separation of the three geographical regions, with the least admixture, was exhibited at $K=3$ (Fig. 2a). One additional cluster was present at $K=4$ and contained 46% NSW, 33% TAS, and 25% NZ samples (Fig. 2b). At $K=5$, genetic clusters relating to the geographical regions were less evident (Fig. 2c).

Initially, outlier SNPs possessed high power to discriminate between individuals from the three regions as 100% of individuals were successfully assigned to their sampling regions with high probabilities (NSW average probability = 0.993 ± 0.02 SD, $n=29$; NZ average probability = 0.986 ± 0.03, $n=23$; TAS average probability = 0.999 ± 0.002, $n=17$) using the self-assignment approach in Geneclass. However, cross-validating the results using the assignment/exclusion approach indicated a reduction in assignment success (Table 4). Successful assignments based on the highest probability of occurrence returned the highest assignment success rates, while 95% probability gave the lowest. This indicates that while assignments based on the highest assignment probability returned good rates of assignment success, there is often less than 5% difference in assignment probability between the most- and second most probable location of occurrence.

Population structuring based on allele sharing methods

No subtle genetic structure was apparent in the ASD NJ dendrogram based on the neutral SNP panel (Fig. 3a); all branches were of similar length, indicating that individuals had equal levels of genetic differentiation, regardless of regional origin. However, weak genetic partitioning was apparent in the NJ dendrogram based on the outlier SNPs (Fig. 3b), with the majority of individuals within a clade originating from the same region (Fig. 3b). Four clades exhibited regional stratification with 73% of NZ individuals belonging to Nodes 2 and 4, 47% of TAS individuals belonging to Node 3, and 31% of NSW individuals belonging to Node 1.

The SAShA analysis of the neutral SNP panel detected very small, but significant, difference between the expected and observed mean distance between shared alleles. Shared alleles were found to be dispersed over broader geographical distances than expected under a model of panmixia (OM = 1,137.6 km; EM = 1132.6 km; $P<0.001$).

Gene flow analysis

The coalescent simulations indicate that the *Sagmariasus verreauxi* breeding stocks located in NSW and NZ consist of a single genetically homogeneous population, as evidenced by the model rankings based on the highest likelihood of occurrence and probability (Table 5). The single genetic population model had a probability 10.6 times higher than that of the next best model of full migration (bi-directional migration from two genetically distinct models). The model found least likely to occur was that of unidirectional gene flow from NZ to NSW.

Relative contemporary migration rates indicated that migration was bi-directionally high between NSW and NZ and between NSW and TAS (Fig. 4). Relative migration rates between NZ and TAS were comparatively lower than the bi-directional rates for NSW. The relative migration rates calculated with Nei's G_{ST} (Fig. 4a) and Jost's D (Fig. 4b) both showed congruence in directionality and magnitude of migration; however, relative migration rates with Jost's D indicated less migration between NZ and TAS than Nei's G_{ST}.

Discussion

In this study, we have used genome-wide ddRADseq data to investigate the broad-scale genetic structure, connectivity, and local adaptation of *Sagmariasus verreauxi* across its known range. For the neutral SNPs, we found a signature of genetic homogeneity that extended across the Tasman Sea, encompassing the NSW and NZ breeding grounds as well as the putative genetic sink population in Tasmania. This result indicated regional disparity in genetic structure of *S. verreauxi* and the closely related *Jasus edwardsii*, for which a genetic break exists between Australian and NZ populations. Due to the genetic homogeneity detected across the three regions, the source population for the Tasmanian recruits could not be identified. We found no evidence to suggest that self-recruitment is occurring in the Tasmanian population; however, we cannot exclude the possibility that self-recruitment in Tasmania is being masked by high gene flow.

Weak regional stratification between NSW, Tasmania, and NZ was evident through the examination of only 15 outlier SNPs. While this result suggests that local adaptation may be occurring, we were unable to associate any of the loci with genes, and therefore, there is a chance that the outliers may be due to Type I error or other demographic processes. Nevertheless, as regional genetic structure with the neutral SNP panel was not detected, we examined the outlier SNP panel to ascertain whether outlier SNPs had increased power to distinguish between regions in a genetically homogeneous species, which has not been examined

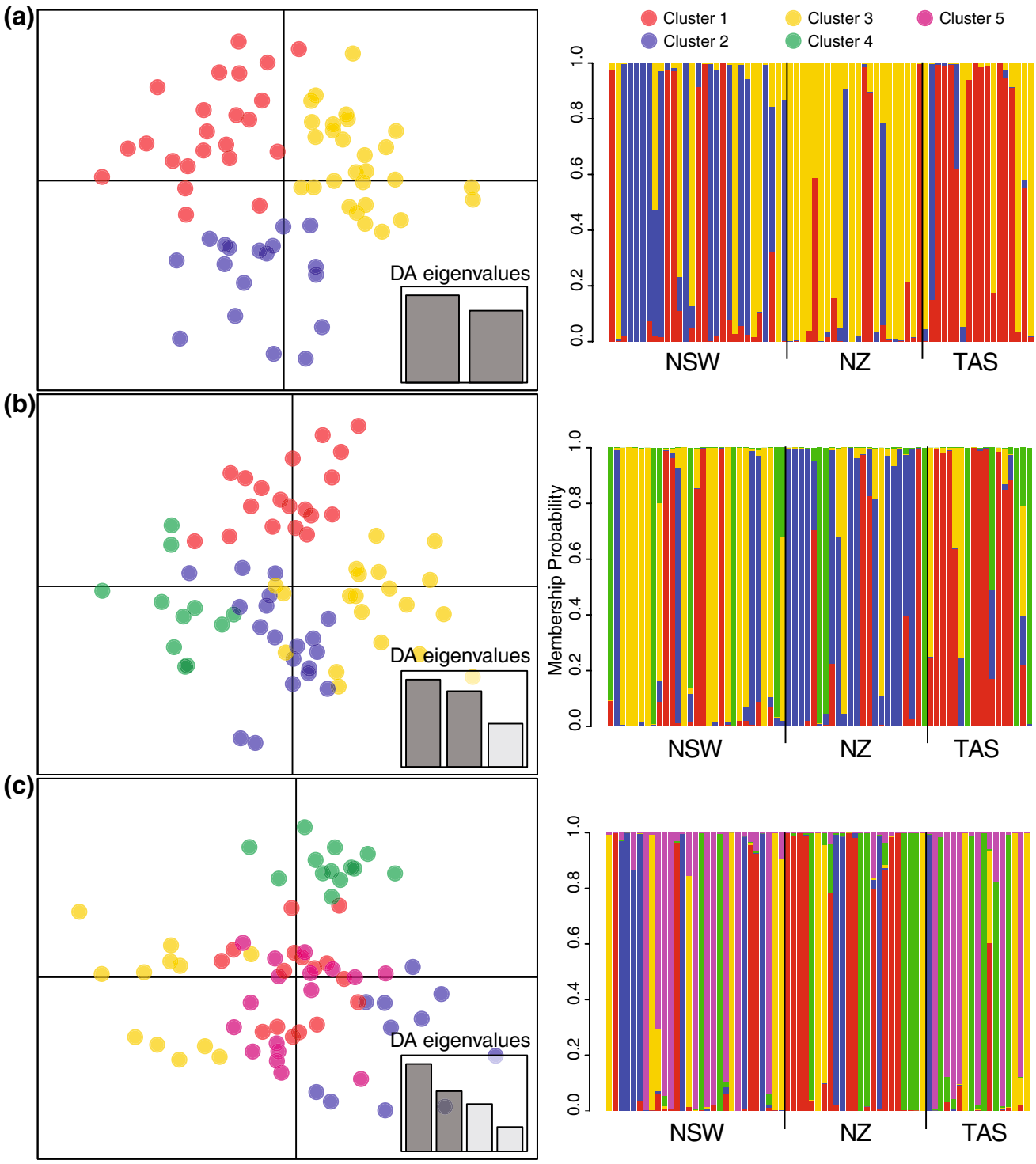

Fig. 2 Discriminate analysis of principal components using the outlier SNP panel at **a** $K=3$, **b** $K=4$, and **c** $K=5$. Scatterplots (on the left) represent individual samples, with the colours indicating cluster assignment. The concordance between actual sampling location (x axis) and probability of cluster assignment based on genotype (y axis) is presented on the right-side panels. Each column represents an individual, and the colours represent the posterior probability of membership to that cluster by the DAPC. *NSW* New South Wales, Australia; *TAS* Tasmania, Australia; *NZ* New Zealand

Table 4 Percentage comparison of successful individual assignments based on (a) the highest probability, (b) probabilities above 0.99, and (c) probabilities above 0.95

	Percentage of individuals successfully assigned to:				
	NSW	NZ	TAS	> 1 population	Unassigned
Successful assignment based on highest probability					
NSW	**71**	7	7	14	0
NZ	64	**18**	9	9	0
TAS	25	0	**63**	12	0
Successful assignment based on > 0.99 probability					
NSW	**43**	0	0	50	7
NZ	9	**18**	9	46	18
TAS	13	13	**38**	25	13
Successful assignment based on > 0.95 probability					
NSW	**29**	0	0	71	0
NZ	9	**9**	9	73	0
TAS	0	0	**13**	75	13

Successful assignments to sampling origin are in bold

before. Outlier SNPs showed good regional assignment success based on the highest probability of occurrence. However, there was often less than 5% probability between the first- and second- most probable locations of occurrence and assignment success rates varied between regions. Nevertheless, this is a promising result and increasing the sampling of SNPs may increase the number of outliers identified and improve regional discrimination in a genetically homogeneous species.

Maintenance of neutral genetic homogeneity

Our results of trans-Tasman neutral genetic homogeneity contradict with the results of Brasher et al. (1992), a preliminary genetic study reporting a genetic break between Australian and NZ *S. verreauxi* populations. This discrepancy in results is most likely due to low ($N = 20$) and uneven sampling effort (NZ $n = 14$, Australian $n = 6$) in the preliminary study, which is known to cause overestimation of genetic stratification calculated via G_{ST} (Caujapé-Castells 2010), the genetic distance estimator used in Brasher et al. (1992). Hence, it is likely that the genetic break detected in Brasher et al. (1992) is an artefact of the sampling effort, not a biological reality.

Lack of neutral genetic structure can occur between discrete regions due to gene flow, or alternatively, due to regions retaining effective population sizes (N_e) large enough

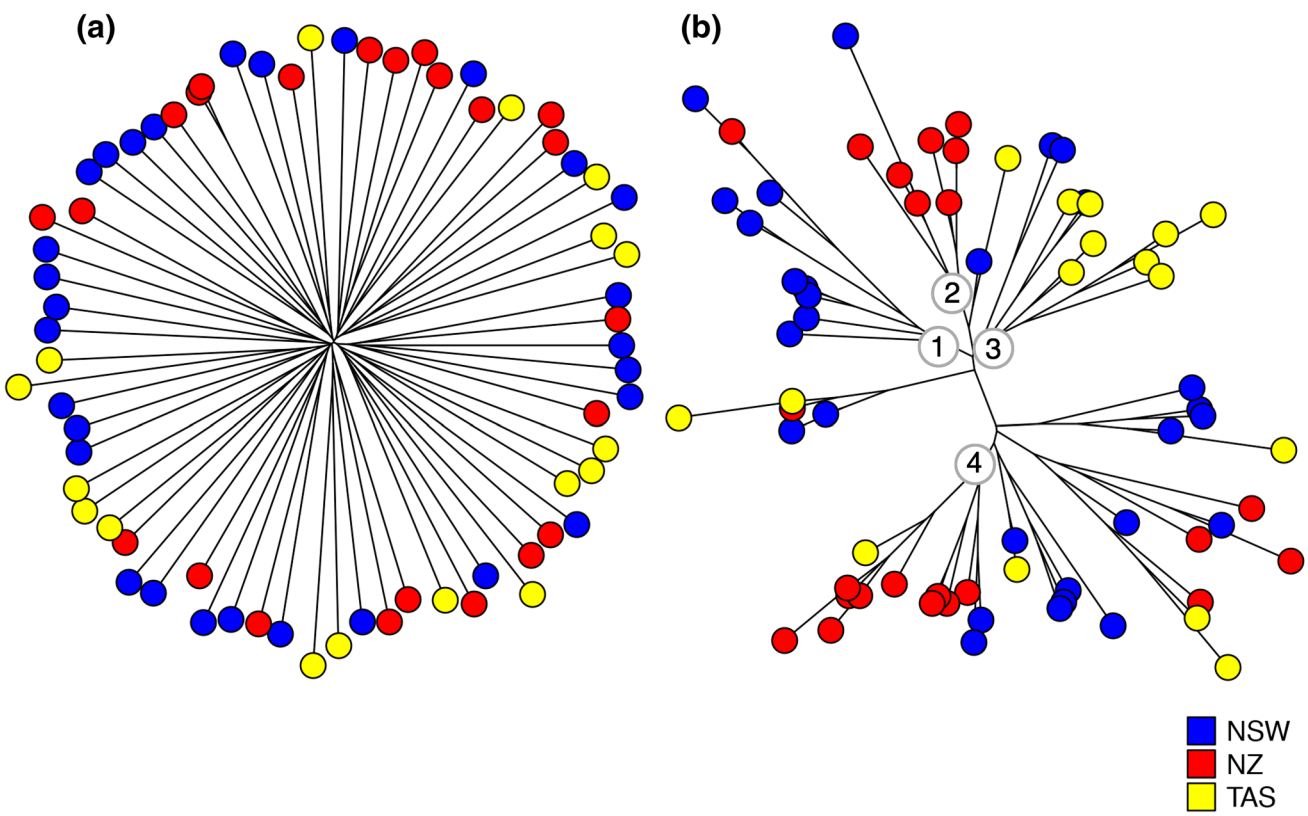

Fig. 3 Neighbour joining dendrogram of **a** neutral SNP panel and **b** outlier SNP panel based on allele sharing distances. Numbers in circles are the node names, as per the text

Table 5 Comparison of log Bayes Factors (LBF) inferred from Bezier approximation score for different models of migration between NSW and NZ breeding populations

Model	Log (mL)	LBF	Rank	Model probability
Single genetic population	−8394.13	0	1	0.9137
Full migration	−8396.49	−2.36	2	0.0862
NSW to NZ only	−8415.84	−21.71	3	3.41×10^{-10}
NZ to NSW only	−8448.69	−54.56	4	1.84×10^{-24}

to overcome the effects of genetic drift, or a combination of both (Hauser and Carvalho 2008). From our study, it is not possible to determine the mechanism responsible for maintaining neutral genetic homogeneity; however, the detection of overly dispersed alleles is consistent with routine long-distance larval dispersal (Kelly et al. 2010). Furthermore, as genetic differentiation is maintained in the closely related Southern Rock Lobster, *J. edwardsii* (Villacorta-Rath et al. 2016) which has a wider distribution, higher biomass, and a longer pelagic larval duration compared to *S. verreauxi*, it is unlikely that genetic homogeneity is being maintained in *S. verreauxi* through a large N_e alone.

Oceanographic modelling supports the hypothesis of genetic homogeneity in *S. verreauxi* being maintained via long-distance larval dispersal. Chiswell et al. (2003) demonstrated that it is theoretically possible for 2% of NSW *S. verreauxi* larvae to reach NZ shores within 12 months if they are entrained in the Tasman Front (see Fig. 1 for Tasman Front details). While our historical gene flow analysis was unable to provide the rate or direction of migration due to the breeding populations consisting of a single genetic stock (i.e., NSW and NZ are genetically identical; therefore, there is no way to statistically discern patterns of gene flow), this result does not preclude gene flow (Horne and Van Herwerden 2013). Our contemporary gene flow analysis exhibited strong bi-directional gene flow between all three regions; however, this is unlikely as there is no evidence that Tasmanian lobsters reproduce. Given that the contemporary gene flow analysis assesses the genetic differentiation between a pool of migrants against two populations, under a scenario where two populations are genetically identical, the pool of migrants will exhibit the same degree of genetic similarity to both populations, and hence, the directionality of gene flow will be deemed to be bi-directional. However, model-based studies have demonstrated that asymmetrical gene flow can cause displacement of the genetic variation within the sink population with that of the source population (Pringle and Wares 2007; Wares and Pringle 2008). Under this scenario, the homogeneous genetic signature produced by asymmetrical gene flow will be indistinguishable from bi-directional gene flow.

Taking into account the local oceanographic features, asymmetrical gene flow from NSW to NZ is the most likely scenario. It is unlikely that trans-Tasman genetic homogeneity is maintained through bi-directional, or unidirectional gene flow originating from NZ, as a strong Tasman Front undercurrent is required for larvae to transit ~ 2000 km

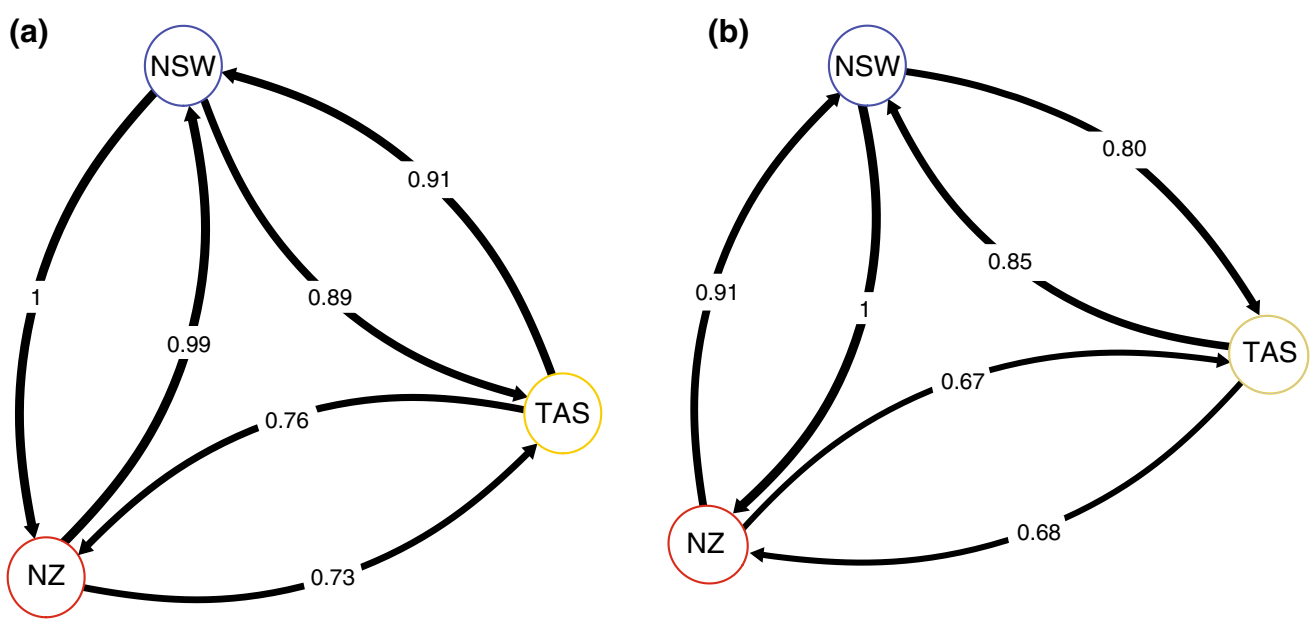

Fig. 4 Relative migration network among all sampling locations, calculated with **a** Nei's G_{ST} and **b** Jost's *D*. The arrows indicate direction of migration, while the numbers and strength of the arrows indicate the relative migration rates. Higher numbers and thicker lines signify stronger relative migration

within their pelagic larval phase. A Tasman Front undercurrent has been identified at a depth of 800 m (Sutton and Bowen 2014); however, this undercurrent is unlikely to be a vector for trans-Tasman spiny lobster dispersal, as it is weak and 400-m deeper than the current known maximum depth of spiny lobster larvae (Rimmer and Phillips 1979; Minami et al. 2001; Bradford et al. 2005). Therefore, we propose unidirectional gene flow from NSW to NZ, facilitated by the Tasman Front, is the most likely scenario for *S. verreauxi*.

The differential population structure observed for *S. verreauxi* and *J. edwardsii* across the Tasman Sea may seem surprising, given their similar life histories; however, patterns of incongruence in regional genetic structure have been found to occur in co-distributed, closely related species of sharks (Ovenden et al. 2009), mackerel (Zardoya et al. 2004), and sea bass (Liu et al. 2006). Explanations for the difference in genetic patterns have been attributed to subtle differences in life history and/or historical environmental changes resulting in differential timing of population isolation. As the Australian and NZ populations of *S. verreauxi* and *J. edwardsii* were estimated to have physically separated approximately 10 MYA (George 1997), this is unlikely to be the cause of the disparity in genetic structure between the two species. More likely is the difference in fecundity of the two species. *Sagmariasus verreauxi* has a maximum fecundity almost four times higher than *J. edwardsii* (Table 1). This combined with the NSW *S. verreauxi* breeding ground being slightly further north than the break off of the Tasman Front could account for more *S. verreauxi* larvae being entrained in the Tasman Front and transported across the Tasman Sea, thus maintaining genetic homogeneity.

While a genetic break exists for *J. edwardsii* across the Tasman Front, the Tasman Front itself is not a barrier to dispersal, as low levels of migration were detected from Australia to New Zealand (Thomas and Bell 2013). The factor that may account for the potential difference of trans-Tasman gene flow between *S. verreauxi* and *J. edwardsii* is the different breeding strategies between the species. *Sagmariasus verreauxi* juveniles undertake a counter current migration to join the geographically restricted breeding grounds with the onset of maturity (Booth 1986), while *J. edwardsii* breed throughout their distribution (Booth and Breen 1994; Booth and Phillips 1994). Therefore, it is possible that *S. verreauxi* larvae are more abundant in an area, where trans-Tasman transport is possible, leading to a higher number of *S. verreauxi* recruits successfully traversing the Tasman Sea than *J. edwardsii* recruits, which could account for the genetic disparity between the species.

Tasmanian recruits

Assignment tests using neutral SNPs were unable to determine the region of origin for the Tasmanian recruits. Tasmanian recruits were just as likely to be assigned back to Tasmania as the other two regions, and due to the absence of a reduction in genetic diversity (i.e., absence of founder effect) in Tasmania, there is no evidence for the occurrence of self-recruitment in Tasmania. Without self-recruitment, the Tasmanian population is likely reliant upon immigrants for its persistence.

While NZ cannot be completely ruled out as a source for Tasmanian recruits, regional oceanography indicates that NSW is the most likely source for Tasmanian recruits due to the southerly flowing East Australian Current (EAC; Ridgway and Hill 2012). The transport of many marine invertebrates from mainland Australia to Tasmania is associated with the EAC, especially recently with the strengthening of the EAC and increased water temperatures that are causing species ranges to shift into Tasmanian waters (Ling et al. 2009; Banks et al. 2010; Johnson et al. 2011; Amor et al. 2014). As NSW is the likely source population, finer scale analysis with higher spatial sampling along the NSW coast should be conducted to examine the existence of isolation by distance or gene surfing patterns, which would indicate range-shifting and self-recruiting (Pierce et al. 2014) in Tasmania. In some cases, fine-scale patterns can be masked at broader scales (Adrian et al. 2017). If this is the case for *S. verreauxi*, self-recruitment within Tasmanian could be obscured by the current regional sampling scale.

Outlier SNPs and their utility in assignment tests

Only a few examples exist in the literature of species exhibiting regionally stratified outlier loci against a background of neutral genetic homogeneity (Gagnaire et al. 2012; Pujolar et al. 2014; Gleason and Burton 2016; Jasonowicz et al. 2017), the cause of which is attributed to occurrence of spatially varying selection from a single gene pool. We were unable to associate any biological functions to the RAD-tags containing outlier SNPs in this study; however, this does not preclude the possibility of spatially varying selection acting through pre- or post-settlement mortality on *S. verreauxi* larvae. Spatially varying selection acts within each cohort; however, it can be stable across cohorts if the environmental conditions remain stable (Gagnaire et al. 2012; Pujolar et al. 2014), creating a regional signal of spatial varying selection, as is the case for *S. verreauxi*. Alternatively, the genetic structure of *S. verreauxi* could result from an absence of drift-migration equilibrium (Slatkin 1993), where diverged populations with restricted gene flow resemble a single genetic population as there has been insufficient time since population divergence for neutral genetic differentiation to occur.

Outlier SNPs show promise for assigning individuals to their region of origin when genetic homogeneity is detected through the analysis of neutral SNPs. Our study

demonstrates that care is needed in deciding the criteria to base successful assignments on, as different criteria greatly influenced the success rates of individual assignments. Using the highest probability returned the best assignment success; however, these results may have been due to chance, as differences between probabilities for each location were low. While the assignment success rates were variable between regions and differences between probabilities were low, these factors could be artefacts of the low number ($n = 15$) of outlier SNPs that were detected in this study. The power of outlier SNPs to assign individuals to their geographical origin is a function of the number of outlier SNPs detected; hence, more outlier SNPs results in higher assignment power (Benestan et al. 2015). As sequencing technology, such as target capture approaches and whole genome sequencing, develops and becomes more accessible, it will be possible to sample more of the genome, providing greater access to highly divergent regions of the genome and outlier SNPS. Even so, with only a small number of SNPs, this study shows that outlier SNPs have the potential to be used for population assignment in species with neutral genetic homogeneity; however, further investigations are needed to determine their reliability as a management tool.

Management implications

This study highlights the complexity of elucidating genetic populations to enhance management strategies for a genetically homogeneous species. Quantifying the direction and magnitude of gene flow is essential to determine the level of population connectivity, and more importantly to resource managers, the demographic connectivity between populations. Demographic independence has been suggested to occur when migration rates between populations fall below 10% (Hastings 1993; Waples and Gaggiotti 2006). However, when a species exhibits genetic homogeneity, migration rates are unable to be statistically determined through genetic data alone. We suggest that, in these cases, oceanographic modelling may be needed to estimate potential migration rates between populations. In the case of *S. verreauxi,* we hypothesise that the genetic homogeneity between NSW and NZ is caused by asymmetrical gene flow; however, biologically realistic (e.g., seeding during egg-hatching months and restricted to the NSW and NZ breeding grounds) oceanographic models are required for further clarity.

Conclusion

Our analyses of neutral genetic variation across the known range of *S. verreauxi* suggest that regional genetic structure is homogeneous at neutral SNPs, indicating the lack of a trans-Tasman genetic break. The genetic disparity between the closely related *S. verreauxi* and *J. edwardsii* indicates the importance of quantifying genetic structure for exploited species rather than using the genetic structure and connectivity of a closely related species as a proxy for management decisions. Outlier SNPs detected weak regional partitioning, which could potentially indicate the presence of spatially varying selection between the regions. Due to the weak regional partitioning of the outlier SNPs, they were useful in achieving high individual assignment success between the regions, albeit with low differentiation between probabilities of first and subsequent population assignments. Although further investigations into the reliability of outlier assignment success are needed, these SNPs do show promise in their utility to be used as traceability tools for species exhibiting genetic homogeneity.

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Data accessibility Reference loci sequences are available through Dryad, https://doi.org/10.5061/dryad.d4q57s5.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed and all necessary approvals were obtained.

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